

ISHITA

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EDUCATION

Bachelor of Technology in Computer Science and Engineering

Expected June 2027

VIT Bhopal University

CGPA: 8.57

Relevant Coursework: Data Structures and Algorithms, Machine Learning, Deep Learning, Database Management Systems, Object-Oriented Programming, Computer Networks, Operating Systems, Statistics for Data Science

Class 12th: 95.6% ; Class 10th: 97.6%

2020 - 2022

Modern Delhi Public School

TECHNICAL SKILLS

Programming Languages: Python, Java, SQL

Machine Learning & AI: TensorFlow, PyTorch, Keras, Scikit-learn, Deep Learning, Neural Networks, CNN, RNN, LSTM, NLP, Computer Vision, Recommender Systems, Supervised & Unsupervised Learning

Data Science: Pandas, NumPy, Matplotlib, Seaborn, Feature Engineering, Data Visualization, EDA

Tools & Technologies: Git, GitHub, Jupyter Notebook, Google Cloud Platform, VS Code, REST APIs, AWS

PROJECTS

DWSIM Surrogate Model for Distillation Column — *Python, Scikit-learn, XGBoost, ANN, DWSIM, Machine Learning*

- Automated DWSIM distillation column simulations using Python, generating a dataset of **500+ operating conditions** via Latin Hypercube Sampling across 7 input variables (feed temperature, pressure, composition, reflux ratio, stages, feed stage, bottoms flow).
- Trained and compared **4 ML models** (Polynomial Regression, Random Forest, XGBoost, ANN) to predict 4 process outputs — distillate/bottoms composition (x_D , x_B) and condenser/reboiler duties (QC , QR).
- Selected **Random Forest** as the final surrogate model with the highest R^2 score and lowest MAE/RMSE across all output variables, achieving **up to 95%+ prediction accuracy** on held-out test data.
- Reduced simulation overhead by replacing repeated DWSIM runs with instantaneous ML inference, enabling rapid process optimization.

Automatic Waste Segregation — *Python, PyTorch, EfficientNet-B0, Flask, Computer Vision*

- Built an end-to-end deep learning waste classification system using EfficientNet-B0 trained on a 15,515-image, 12-class garbage dataset with data augmentation (flips, rotation, ColorJitter).
- Deployed a Flask web application with drag-and-drop image upload providing real-time predictions, confidence scores, recyclability status, and bin disposal guidance.
- Achieved **73.10% validation accuracy** across 12 waste categories including cardboard, plastic, glass, battery, and biological waste.
- Split dataset into **75% train / 15% validation / 10% test** sets using a fixed random seed (42) for reproducibility, and used AdamW optimizer with CosineAnnealingLR scheduler trained over **20 epochs**.

CERTIFICATIONS

Applied Machine Learning in Python — University of Michigan

Dec 2024

Workshop on Containers and AIML on AWS Cloud — Cognizance, IIT Roorkee

Apr 2024

AWS Cloud Clubs Machine Learning Camper — United Latino Students Association

Mar 2024

Crash Course on Python — Google

Nov 2023

ADDITIONAL INFORMATION

- **Languages:** Fluent in English and Hindi; Basic proficiency in French
- **Achievement:** Cleared two competitive rounds and reached the final stage of a Health Hackathon by developing an ML-powered healthcare solution; applied machine learning algorithms and data analytics to build an innovative system recognized by industry experts and judges.
- **Soft Skills:** Problem-solving, teamwork, leadership, communication, time management, adaptability